

untangle::actuator

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Chapter 1

actuator

An actuator is a generic callable that may invoke a list of actions(callables of type `std::function<...>`).

Its goal is to provide a light and simple mechanism to connect components that want to communicate with each other. It works similar as the "signal and slots" mechanisms from other frameworks, but much simplified.

Modern programming languages do not offer class intrinsic interfaces that would improve significantly how objects are connected and how they communicate. Instead, languages, like C++, are using external interfaces, that are not only cluttering the code, but they don't feel "natural" as they are opposed to what an interface is in reality. Basically the objects and interfaces are not separated, and the interface is intrinsic to the object(think on the cogs in a mechanism). A generic callable is an attempt to mitigate this problem. `async` implementation is such use case.

An actuator provides the results of all actions in a vector, in the same order as the actions were added.

Example: how to use actuator instead of polymorphism

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle)
{
    for (const auto& s : shapes)
    {
        s->rotate(angle);
    }
}

std::shared_ptr<triangle> t(new triangle);
std::shared_ptr<circle> c(new circle);
std::shared_ptr<square> s(new square);

// using polymorphism
std::vector<shape*> shapes;
shapes.push_back(t.get());
shapes.push_back(c.get());
shapes.push_back(s.get());

rotate_shapes(shapes, 10);

// using actuator
auto action1 = untangle::bind(t, &triangle::rotate);
auto action2 = untangle::bind(c, &circle::rotate);
auto action3 = untangle::bind(s, &square::rotate);
auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

For convenience there are provided helpers methods to "connect" to an initial list of "actions", or to create bindings to class methods.

Please check the manual in [doc/refman.pdf](#) for further references.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [11](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::actuator functor	15
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

untangle::actuator< action_t >	
An actuator is a functor that can trigger a dynamic list of actions (of type std::↵ function<...>)	17
untangle::invalid_action	
Invalid action exception	24

Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

actuator.hpp	27
--	----

Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Functions

- `template<typename action_t, typename... Actions>`
`actuator< action_t > untangle::connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↔`
`T>::type>>`
`action_t untangle::bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↔`
`T>::type>>`
`action_t untangle::bind (class_t *obj, T class_t::*method)`
Binding to a class method.

6.1.1 Detailed Description

6.1.2 Function Documentation

6.1.2.1 connect()

```
template<typename action_t, typename... Actions>
actuator< action_t > untangle::connect (
    action_t & A1,
    Actions &... An)
```

Creates an actuator holding an initial list of actions.

Parameters

A1	- The first action. It is specified as <code>std::function<...></code> .
----	--

An	- Any number of further actions, of the same type as A1.
----	--

Returns

An [actuator](#).

Example:

```
void rotate_shapes(const std::vector<shape*>& shapes, int angle) {
    for (const auto& s : shapes) {
        s->rotate(angle);
    }
}

auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate(20);
```

6.1.2.2 bind() [1/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::<
type>>>
action_t untangle::bind (
    const std::shared_ptr< class_t > & obj,
    T class_t::* method)
```

Binding to a class function member.

It returns a `std::function(lambda)` that wraps the function member. It may be used to provide an action for [connect\(\)](#) or [actuator::add\(\)](#).

Remarks

It requires a shared pointer to the class type. A `std::weak_ptr` to it is captured internally in a lambda, so the binding can always check whether the shared object is still alive without keeping it alive itself. Therefore, it is safe to use actions provided by this binding inside an [actuator](#), and safe for the action to outlive the caller's shared pointer.

Parameters

obj	- Class object.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

`action_t` - A `std::function` that wraps the pointer to function member.

Remarks

If the class object gets invalid, invoking this binding will throw an exception of type [invalid_action](#).

6.1.2.3 bind() [2/2]

```
template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<T>::↵  
type>>  
action_t untangle::bind (  
    class_t * obj,  
    T class_t::* method)
```

Binding to a class method.

Attention

It is not safe to use this binding when the pointed-to object may be destroyed. A null pointer is detected and reported as a dead action, but a pointer to an already destroyed object can not be distinguished from a valid one. Prefer the overload taking a `std::shared_ptr`.

Remarks

It is provided for convenience of use: within a class it is safe to create bindings through this pointer. Binding a null pointer produces an action that is always dead: invoking it throws [invalid_action](#), and an [actuator](#) drops it.

Parameters

obj	- Pointer to class.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code>

Returns

action_t - A `std::function` that wraps the pointer to function member.

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to [untangle::actuator](#) functor.

Classes

- struct [invalid_action](#)
Invalid action exception.
- struct [actuator](#)
An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Functions

- `template<typename... Args>`
`auto last_arg (Args &&... args)`
Get last argument of params pack;.
- `template<typename action_t, typename... Actions>`
`actuator< action_t > connect (action_t &A1, Actions &... An)`
Creates an actuator holding an initial list of actions.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (const std::shared_ptr< class_t > &obj, T class_t::*method)`
Binding to a class function member.
- `template<typename class_t, typename T, typename action_t = std::function<typename function_remove_const<↵`
`T>::type>>`
`action_t bind (class_t *obj, T class_t::*method)`
Binding to a class method.

7.1.1 Detailed Description

Interface to [untangle::actuator](#) functor.

7.1.2 Function Documentation

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7.1.2.1 last_arg()

Template Parameters

Args	params template types
------	-----------------------

Parameters

args	params
------	--------

Returns

last param value in case it exist, otherwise 0.

Chapter 8

Class Documentation

8.1 untangle::actuator< action_t > Struct Template Reference

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

```
#include <actuator.hpp>
```

Public Types

- using `actions_t` = `std::list<action_t*>`
Actions container type.
- using `results_t` = `std::vector<typename result_t::type>`
Results container type.

Public Member Functions

- `actuator & operator=` (`const actuator &other`)=default
Assignment operator.
- `action_t type` () const
Helper method to access the action type of this object.
- void `reset` ()
Remove all actions and any stored results.
- `template<typename last_t>`
void `invoke_callback` (`last_t &last`)
Invoke the trailing callback argument, if the invocation has one.
- `template<typename... Args>`
void `operator()` (`Args &&... args`)
The call operator.
- `template<typename... Args>`
void `invoke_action` (`const std::string &name, Args &&... args`)
Invokes one single action associated with a key.
- void `add` (`action_t *action`)
Add an action to the actions list.
- void `add` (`const std::string &name, action_t *action`)
Add action to the actions map associated with a name.

- void [remove](#) (const action_t *action)
Remove an action from the actions list.
- void [remove](#) (const std::string &name)
Remove an action from actions map.
- bool [is_connected](#) () const
Check if this actuator is "connected" with other actions.
- bool [has_action](#) (const std::string &name) const
Check if there is certain named action.

Public Attributes

- [actions_t](#) actions
Actions list.
- actions_map_t actions_map
Named actions map.
- [results_t](#) results
Actions return values list.

8.1.1 Detailed Description

```
template<typename action_t>
struct untangle::actuator< action_t >
```

An actuator is a functor that can trigger a dynamic list of actions (of type `std::function<...>`).

Remarks

An actuator object can be constructed with an initial list of actions by [connect\(\)](#).

Callback convention: when the last argument of an invocation is callable with the action return type R, it is not passed on as a plain argument alone – it is also treated as a completion callback. The action is invoked with the full argument list as usual, and the callback is then invoked with the action return value. Any callable qualifies: a `std::function<void(R)>`, a lambda taking R, a function pointer. It applies to [operator\(\)\(\)](#) and to [invoke_action\(\)](#) alike. Actions returning void have no result to report, so no callback is invoked for them.

Warning

The convention keys on the argument being callable, so an action whose last parameter is a callable it means to consume as data – a comparator, a factory – gets it invoked as a callback too, as long as it accepts R. Such an action should not take that parameter last.

Template Parameters

action_t	Action type. It is specified as <code>std::function<...></code> .
----------	---

8.1.2 Member Typedef Documentation

8.1.2.1 actions_t

```
template<typename action_t>
using untangle::actuator< action_t >::actions_t = std::list<action_t*>
```

Actions container type.

Remarks

The elements stored are of pointer type, that is required to implement the [remove\(\)](#) operation. `std::function` supports only equality operator for nullptr (two `std::function(s)` can not compare).

8.1.2.2 results_t

```
template<typename action_t>
using untangle::actuator< action_t >::results_t = std::vector<typename result_t::type>
```

Results container type.

It holds the return values of the actions that have a non-void return type. Upon the actuator invocation, the returns can be extracted from [results](#).

8.1.3 Member Function Documentation

8.1.3.1 operator=()

```
template<typename action_t>
actuator & untangle::actuator< action_t >::operator= (
    const actuator< action_t > & other) [default]
```

Assignment operator.

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
untangle::actuator<decltype(actuator_rotate.type())> actuator_rotate_1 = actuator_rotate;
actuator_rotate_1(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.2 type()

```
template<typename action_t>
action_t untangle::actuator< action_t >::type () const [inline]
```

Helper method to access the action type of this object.

Note

Intended to be used in expressions by decltype(<[actuator](#) instance>.[type\(\)](#))

Returns

action_t

8.1.3.3 reset()

```
template<typename action_t>
void untangle::actuator< action_t >::reset () [inline]
```

Remove all actions and any stored results.

After this call the actuator is empty: [actuator::is_connected\(\)](#) returns false.

8.1.3.4 invoke_callback()

```
template<typename action_t>
template<typename last_t>
void untangle::actuator< action_t >::invoke_callback (
    last_t & last) [inline]
```

Invoke the trailing callback argument, if the invocation has one.

Does nothing unless last_t can be called with the action return type, so any other trailing argument is left alone. The test is on what the type can do, not on what it is, so a plain lambda qualifies as well as a std::function. See the callback convention on [actuator](#).

Remarks

The caller must copy the callback out of the argument pack before invoking the action: an action taking it by value moves from the caller's std::function, which would leave this an empty function to call.

Template Parameters

last↔ _t	Type of the last argument, as returned by last_arg() .
-------------	--

Parameters

last	- Last argument value, copied before the action was invoked.
------	--

8.1.3.5 operator>()()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::operator() (
    Args &&... args) [inline]
```

The call operator.

Actions in the [actuator::actions](#) list are triggered by invoking the call operator.

Parameters

args	- Arguments list must match the action arity. A trailing callback is invoked once per action, with that action's return value; see the convention on actuator .
------	---

8.1.3.6 invoke_action()

```
template<typename action_t>
template<typename... Args>
void untangle::actuator< action_t >::invoke_action (
    const std::string & name,
    Args &&... args) [inline]
```

Invokes one single action associated with a key.

Parameters

name	- Key associated with the action.
args	- Arguments list must match the action arity. A trailing callback is invoked with the action's return value; see the convention on actuator .

8.1.3.7 add() [1/2]

```
template<typename action_t>
void untangle::actuator< action_t >::add (
    action_t * action) [inline]
```

Add an action to the actions list.

Parameters

action	- Action to be added.
--------	-----------------------

Example:

```

const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(2).WillRepeatedly(testing::Return());
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.add(&action1);
actuator_rotate(20);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());

```

8.1.3.8 add() [2/2]

```

template<typename action_t>
void untangle::actuator< action_t >::add (
    const std::string & name,
    action_t * action) [inline]

```

Add action to the actions map associated with a name.

Parameters

name	- Name of the action.
action	- Action to be added.

8.1.3.9 remove() [1/2]

```

template<typename action_t>
void untangle::actuator< action_t >::remove (
    const action_t * action) [inline]

```

Remove an action from the actions list.

An invalid action (empty std::function) is implicitly removed when `operator()()` is invoked.

Parameters

action	- Action to be removed.
--------	-------------------------

Example:

```
const auto t = std::make_shared<triangle_mock>();
const auto c = std::make_shared<circle_mock>();
const auto s = std::make_shared<square_mock>();

EXPECT_CALL(*t, rotate(testing::_)).Times(0);
EXPECT_CALL(*c, rotate(testing::_)).WillOnce(testing::Return());
EXPECT_CALL(*s, rotate(testing::_)).WillOnce(testing::Return());
auto action1 = untangle::bind(t, &triangle_mock::rotate);
auto action2 = untangle::bind(c, &circle_mock::rotate);
auto action3 = untangle::bind(s, &square_mock::rotate);

auto actuator_rotate = untangle::connect(action1, action2, action3);
actuator_rotate.remove(&action1);
actuator_rotate(50);

testing::Mock::VerifyAndClearExpectations(t.get());
testing::Mock::VerifyAndClearExpectations(c.get());
testing::Mock::VerifyAndClearExpectations(s.get());
```

8.1.3.10 remove() [2/2]

```
template<typename action_t>
void untangle::actuator< action_t >::remove (
    const std::string & name) [inline]
```

Remove an action from actions map.

Parameters

name	- Name of the action to remove.
------	---------------------------------

8.1.3.11 is_connected()

```
template<typename action_t>
bool untangle::actuator< action_t >::is_connected () const [inline]
```

Check if this actuator is "connected" with other actions.

Returns

- true - if the `actuator::actions` list is not empty.
- false - if the `actuator::actions` list is empty.

8.1.3.12 has_action()

```
template<typename action_t>
bool untangle::actuator< action_t >::has_action (
    const std::string & name) const [inline]
```

Check if there is certain named action.

Parameters

name	- Name associated with the action.
------	------------------------------------

Returns

- true - if name can be found in [actuator::actions_map](#)
- false - if name can not be found in [actuator::actions_map](#)

The documentation for this struct was generated from the following file:

- [actuator.hpp](#)

8.2 untangle::invalid_action Struct Reference

Invalid action exception.

```
#include <actuator.hpp>
```

Public Member Functions

- [invalid_action](#) (std::string text)
Construct a new invalid action object.
- const char * [what](#) () const noexcept override
The message text describing the reason of this exception.

8.2.1 Detailed Description

Invalid action exception.

Remarks

An action may be provided as a binding to a class function member, by using [untangle::bind\(\)](#). When the class object gets invalid, invoking the action will raise an exception to this type.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 invalid_action()

```
untangle::invalid_action::invalid_action (  
    std::string text) [inline], [explicit]
```

Construct a new invalid action object.

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.2.3 Member Function Documentation

8.2.3.1 what()

const char * untangle::invalid_action::what () const [inline], [override], [noexcept]

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- actuator.hpp

Chapter 9

File Documentation

9.1 actuator.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <exception>
00009 #include <functional>
00010 #include <iostream>
00011 #include <list>
00012 #include <map>
00013 #include <memory>
00014 #include <string>
00015 #include <type_traits>
00016 #include <utility>
00017 #include <vector>
00018
00019 namespace untangle {
00020 // exception
00029 struct invalid_action : std::exception {
00035     explicit invalid_action(std::string text) : message(std::move(text)) {}
00036
00042     const char* what() const noexcept override { return message.c_str(); }
00043
00044 private:
00045     std::string message;
00046 };
00047
00055 template<typename... Args>
00056 auto last_arg(Args&&... args) {
00057     if constexpr (sizeof...(Args) <= 0) {
00058         return 0;
00059     } else {
00060         return std::get<sizeof...(Args)-1>(std::forward_as_tuple(args...));
00061     }
00062 }
00063
00085 template <typename action_t>
00086 struct actuator final {
00094     using actions_t = std::list<action_t*>;
00095     using actions_map_t = std::map<std::string, action_t*>;
00096     using result_t = std::conditional<std::is_void<typename action_t::result_type>::value, int,
00097                                     typename action_t::result_type>;
00104     using results_t = std::vector<typename result_t::type>;
00105
00106     actions_t actions;
00107     actions_map_t actions_map;
00108     results_t results;
00109
00110     actuator() = default;
00111     actuator(const actuator&) = default;
00112     actuator(actuator&&) noexcept = default;
00113     ~actuator() = default;
00120     actuator& operator=(const actuator& other) = default;
00121     actuator& operator=(actuator&&) noexcept = default;
00122
00130     action_t type() const { return nullptr; }
00131
00137     void reset() {
```

```

00138     actions.clear();
00139     actions_map.clear();
00140     results.clear();
00141 }
00142
00158 template <typename last_t>
00159 void invoke_callback(last_t& last) {
00160     if constexpr (std::is_invocable_r_v<void, last_t&, typename result_t::type>) {
00161         last(results.back());
00162     }
00163 }
00164
00173 template <typename... Args>
00174 void operator()(Args&&... args) {
00175     results.clear();
00176
00177     // Copied up front, before any action can move the callback out of the argument pack.
00178     auto last = last_arg(args...);
00179
00180     // Dead bindings are collected here and dropped after the loop.
00181     // The actuator does not own the actions it points at, so it must never
00182     // write through action_t* into a std::function belonging to the caller.
00183     std::vector<action_t*> dead_actions;
00184
00185     for (const auto& action : actions) {
00186         // A null pointer or an empty std::function can never be invoked. Calling an
00187         // empty one throws std::bad_function_call, which is not an invalid_action and
00188         // would escape this operator, so drop it instead of invoking it.
00189         if (action == nullptr || !*action) {
00190             dead_actions.push_back(action);
00191             continue;
00192         }
00193
00194         try {
00195             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00196                 (*action)(std::forward<Args>(args)...);
00197             } else {
00198                 results.push_back((*action)(std::forward<Args>(args)...));
00199                 invoke_callback(last);
00200             }
00201         } catch (const invalid_action& ia) {
00202             std::cout << ia.what() << std::endl;
00203             dead_actions.push_back(action);
00204         }
00205     }
00206
00207     for (const auto& dead_action : dead_actions) {
00208         actions.remove(dead_action);
00209     }
00210 }
00211
00219 template <typename... Args>
00220 void invoke_action(const std::string& name, Args&&... args) {
00221     results.clear();
00222
00223     // Copied up front, before the action can move the callback out of the argument pack.
00224     auto last = last_arg(args...);
00225     const auto& it = actions_map.find(name);
00226     if (it != actions_map.end()) {
00227         try {
00228             if constexpr (std::is_same_v<typename action_t::result_type, void>) {
00229                 (*it->second)(std::forward<Args>(args)...);
00230             } else {
00231                 results.push_back((*it->second)(std::forward<Args>(args)...));
00232                 invoke_callback(last);
00233             }
00234         } catch (const invalid_action& ia) {
00235             std::cout << ia.what() << std::endl;
00236             actions_map.erase(name);
00237         }
00238     }
00239 }
00240
00249 void add(action_t* action) { actions.push_back(action); }
00250
00257 void add(const std::string& name, action_t* action) { actions_map.emplace(name, action); }
00258
00269 void remove(const action_t* action) {
00270     actions.remove_if([&action](const auto& a) { return (action == a); });
00271 }
00272
00278 void remove(const std::string& name) { actions_map.erase(name); }
00279
00286 bool is_connected() const { return !actions.empty() || !actions_map.empty(); }
00287
00295 bool has_action(const std::string& name) const {
00296     return actions_map.find(name) != actions_map.end();

```

```

00297 }
00298 };
00299
00314 template <typename action_t, typename... Actions>
00315 actuator<action_t> connect(action_t& A1, Actions&... An) {
00316     using actuator_t = untangle::actuator<action_t>;
00317     actuator_t actuator;
00318     actuator.actions = {&A1, &An...};
00319
00320     // remove empty actions
00321     actuator.actions.remove_if([](const auto& action) { return (*action == nullptr); });
00322     return actuator;
00323 }
00324
00325 template <typename actuator_t>
00326 void remove_empty_actions(actuator_t& actuator) {
00327     std::vector<typename actuator_t::actions_map_t::const_iterator> removable_iterators;
00328     for (auto it = actuator.actions_map.begin(); it != actuator.actions_map.end(); ++it) {
00329         if (it->second == nullptr) {
00330             removable_iterators.push_back(it);
00331         }
00332     }
00333     for (auto it : removable_iterators) {
00334         actuator.actions_map.erase(it);
00335     }
00336 }
00337
00338 template <typename key_t, typename action_t, typename... Actions>
00339 actuator<action_t> connect(std::pair<key_t, action_t*> A1, Actions... An) {
00340     using actuator_t = untangle::actuator<action_t>;
00341     actuator_t actuator;
00342     actuator.actions_map = {A1, An...};
00343
00344     // remove empty actions
00345     remove_empty_actions(actuator);
00346     return actuator;
00347 }
00348
00349 // generic helpers to remove const qualifier from a function type,
00350 // for instance const member functions
00351 template <typename T>
00352 struct function_remove_const;
00353
00354 template <typename R, typename... Args>
00355 struct function_remove_const<R(Args...)> {
00356     using type = R(Args...);
00357 };
00358
00359 template <typename R, typename... Args>
00360 struct function_remove_const<R(Args...) const> {
00361     using type = R(Args...);
00362 };
00363
00364
00365 template <typename class_t, typename T,
00366         typename action_t = std::function<typename function_remove_const<T>::type>
00367         action_t bind(const std::shared_ptr<class_t>& obj, T class_t::* method) {
00368     return [wp = std::weak_ptr<class_t>(obj), method](auto&&... args) ->
00369         typename action_t::result_type {
00370         // lock() also keeps the object alive for the duration of the call
00371         const auto obj_ = wp.lock();
00372         if (!obj_) {
00373             // inform the actuator about dead binding
00374             throw invalid_action("bind: invalid object");
00375         }
00376         return ((*obj_).*method)(std::forward<decltype(args)>(args)...);
00377     };
00378 }
00379
00380 template <typename class_t, typename T,
00381         typename action_t = std::function<typename function_remove_const<T>::type>
00382         action_t bind(class_t* obj, T class_t::* method) {
00383     return [obj, method](auto&&... args) -> typename action_t::result_type {
00384         if (!obj) {
00385             // inform the actuator about dead binding
00386             throw invalid_action("bind: invalid object");
00387         }
00388         return ((obj)->*method)(std::forward<decltype(args)>(args)...);
00389     };
00390 }
00391
00392 } // namespace untangle

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